

/// Brain Tumor Detection ///

Presented by

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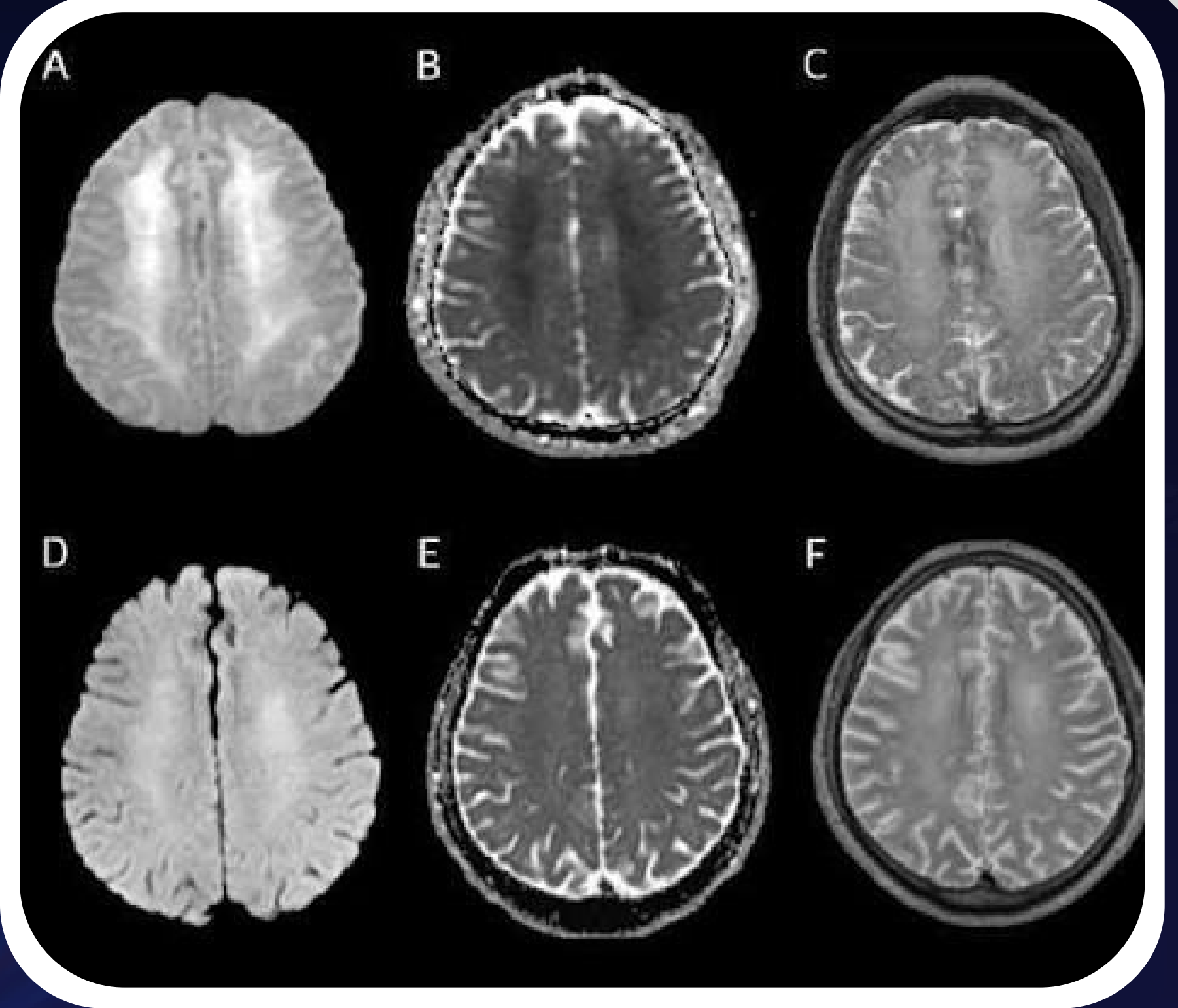
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Project Overview

Our project is an AI-powered brain tumor analysis system. Using MRI images, the system helps detect and classify brain tumors, locate tumor regions, provide medical recommendations, assist users through an AI chatbot, and generate detailed medical reports



Problem Statement



Expert Radiologists

Requires experienced specialists.



Time-Consuming

Deploy the system on cloud platforms.



Tumor Classification

Difficult to classify tumors accurately.



Tumor Localization

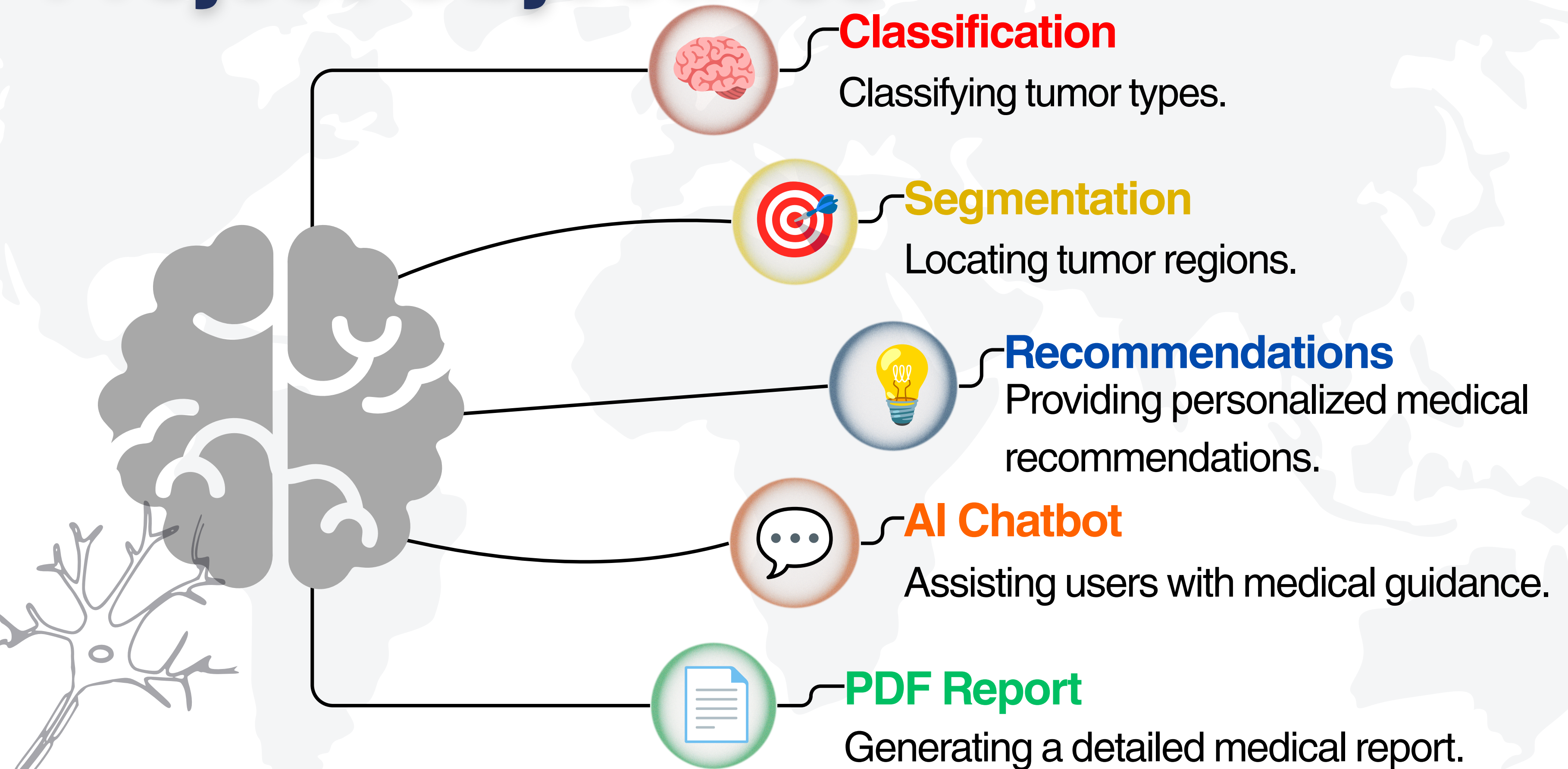
Precise localization is challenging.



Delayed Diagnosis

May affect treatment decisions.

Project Objectives



Dataset & Data Balancing

- **4 CLASSES**

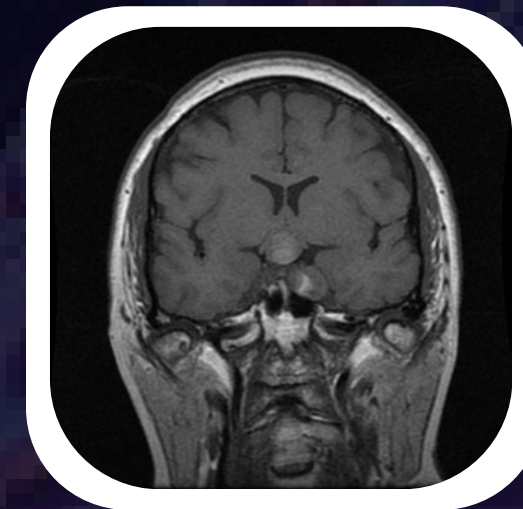
Glioma



Meningioma



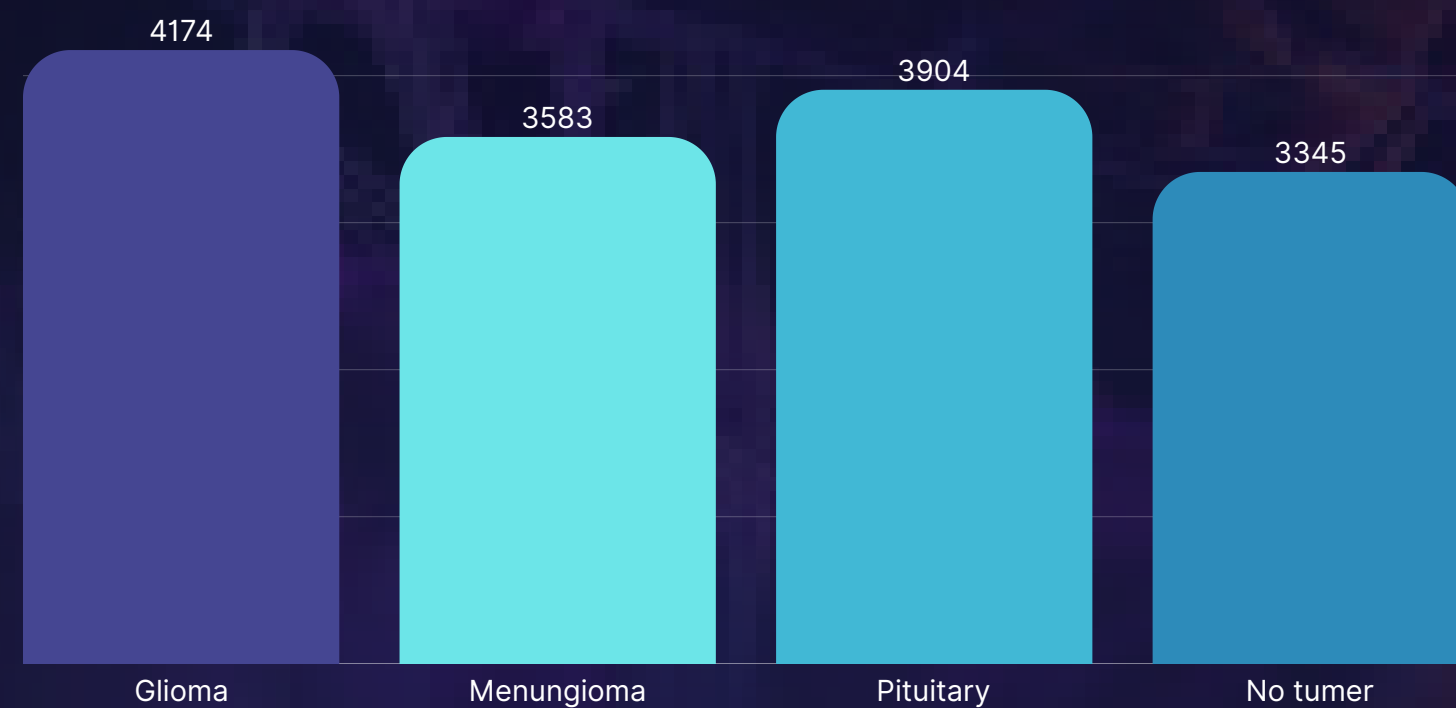
Pituitary



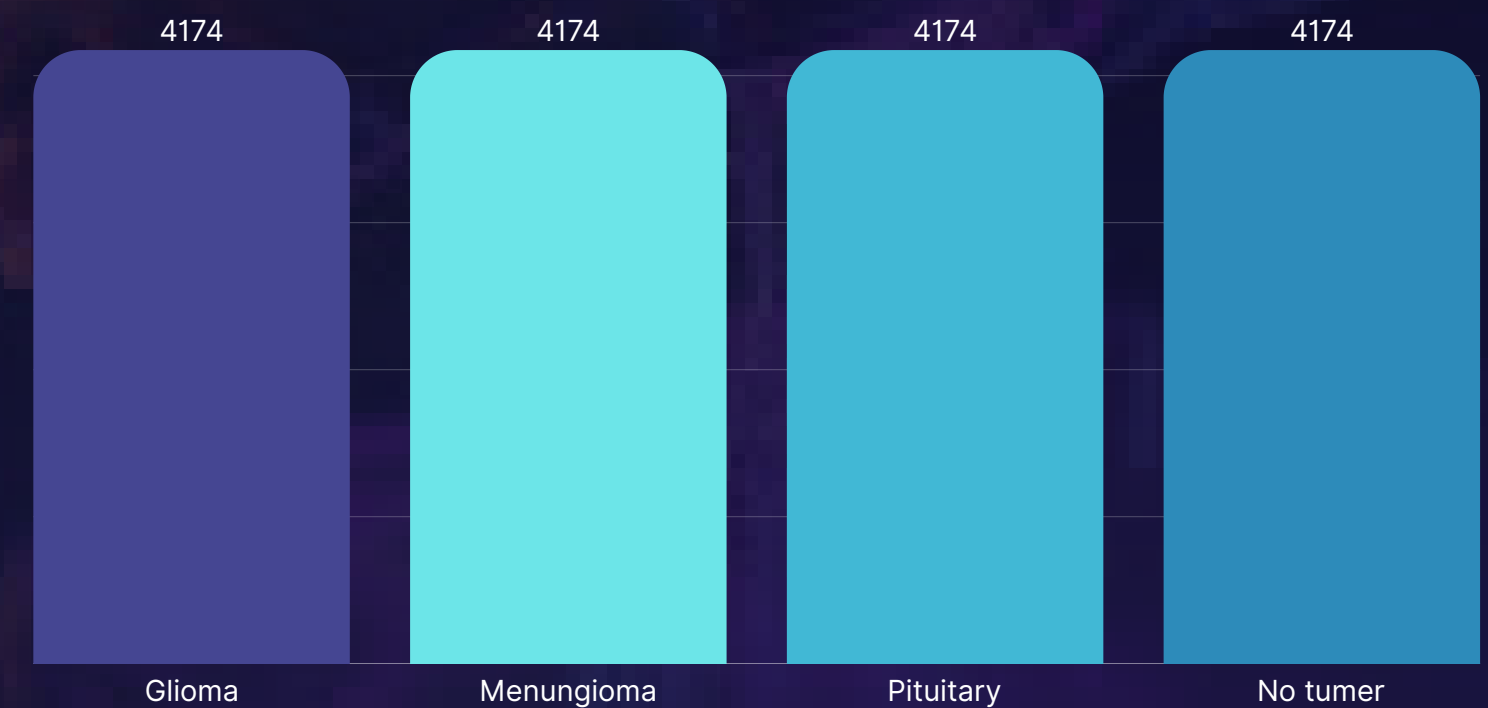
No Tumor



Before Balancing
Total: 15,006

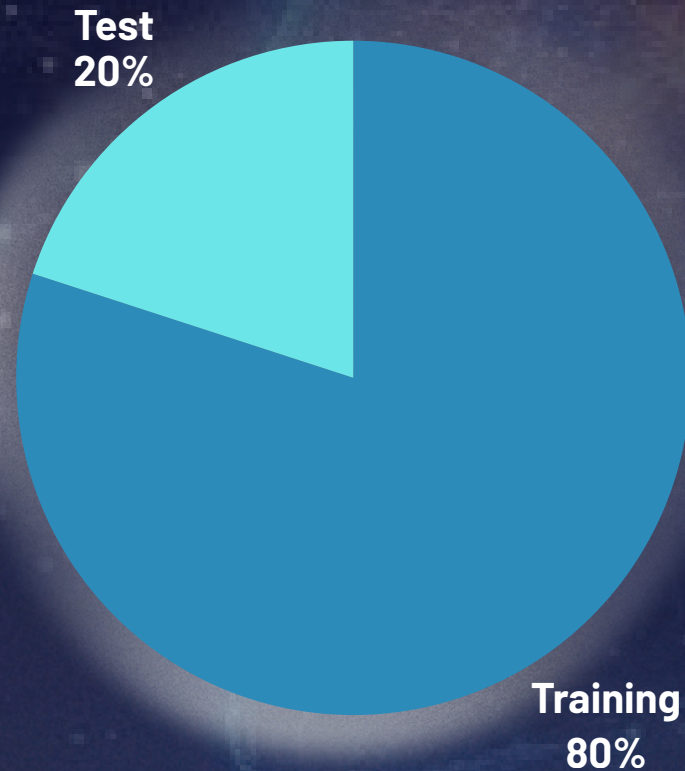


After Balancing
Total: 16,696



Model Training

Data Split:



Dataset Preparation:

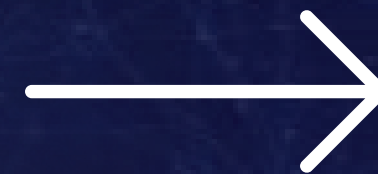
1. Data Cleaning
2. Image Resizing
3. Data Augmentation
4. 80/20 Train-Test Split

Models:

CNN



ResNet50



VGG16

CNN Model

Training Started

Epoch 1/20

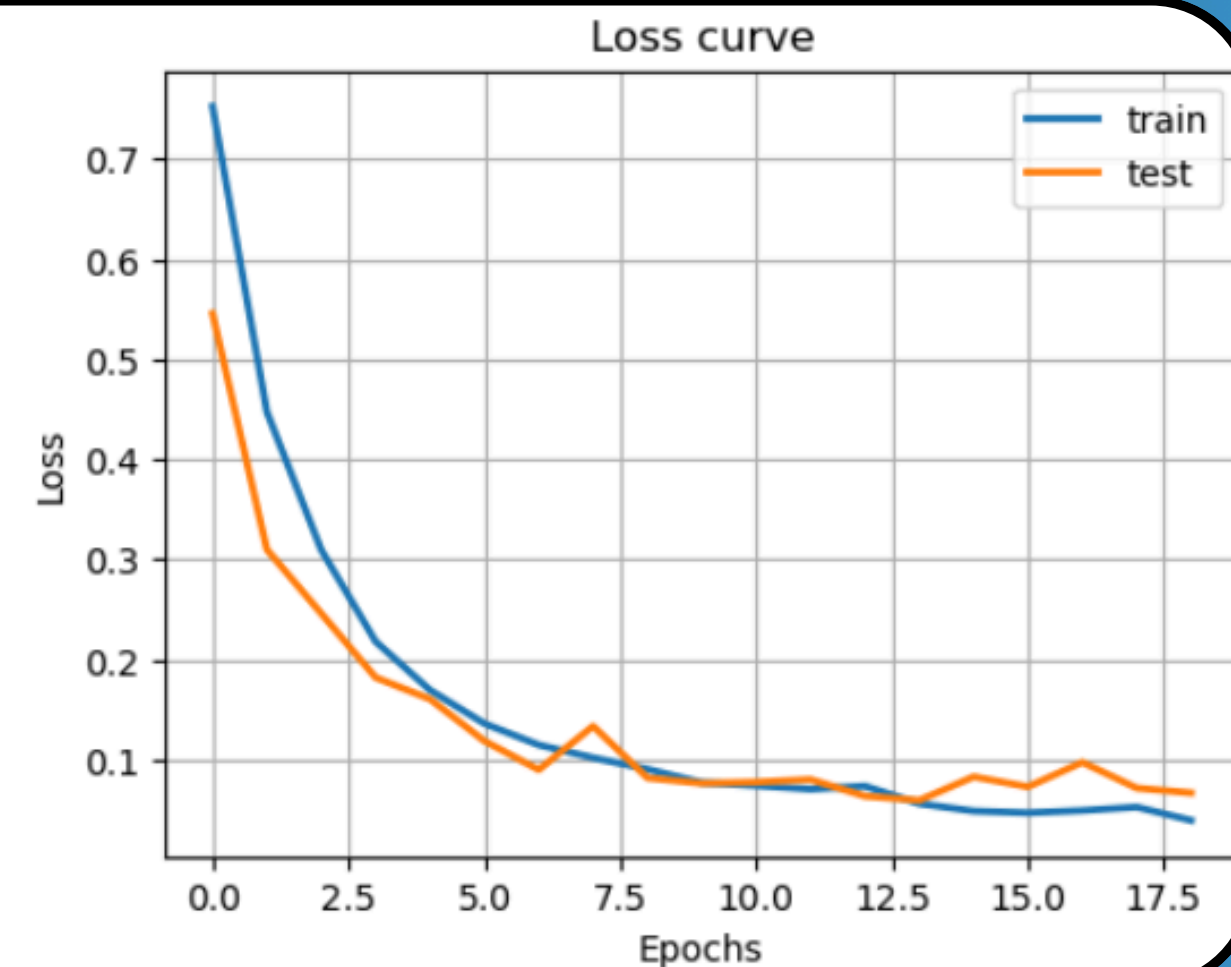
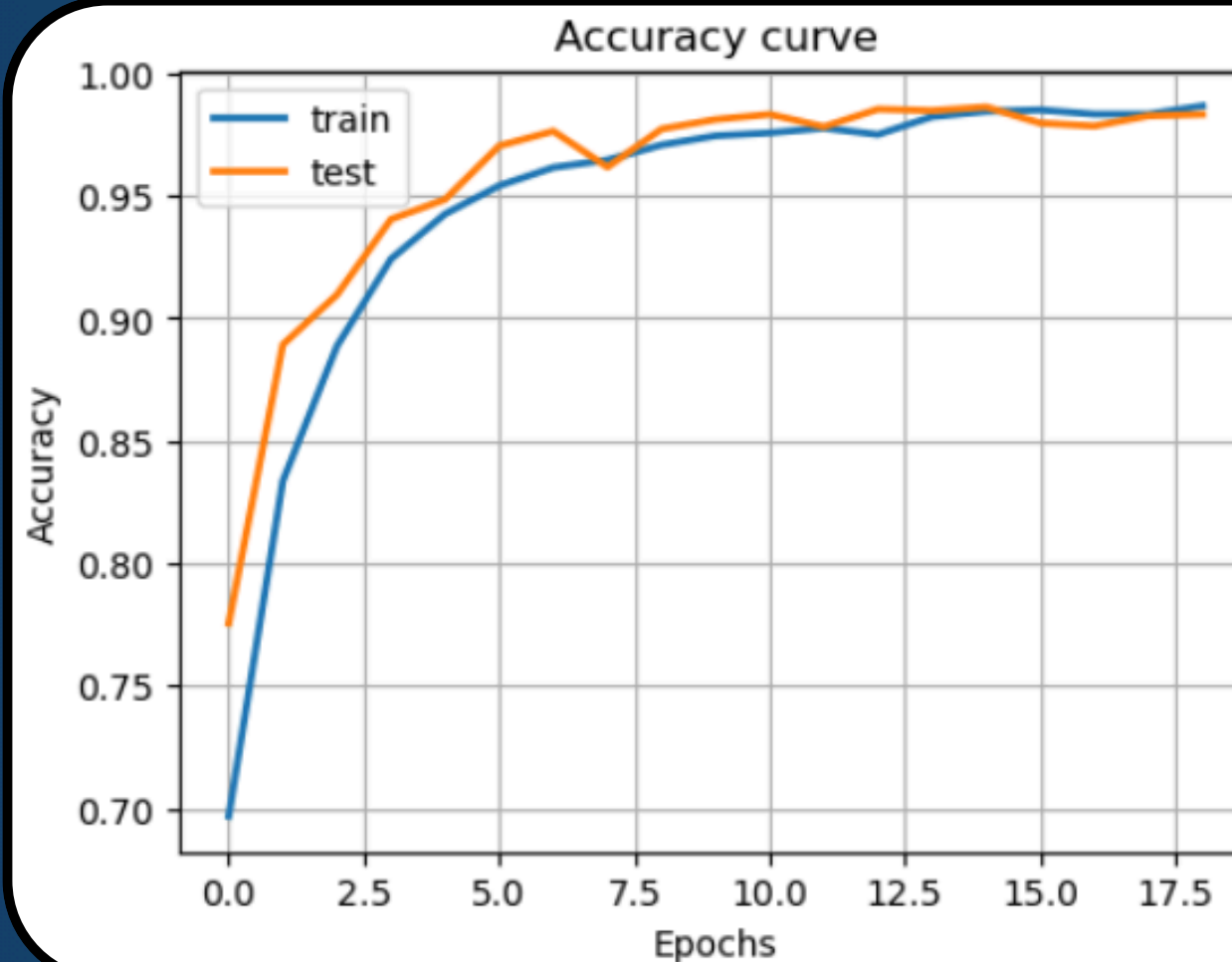
418/418 ————— 1011s 2s/step - accuracy: 0.5989 - loss: 0.9439 - val_accuracy: 0.7754 - val_loss: 0.5454

Training Completed

Epoch 19/20

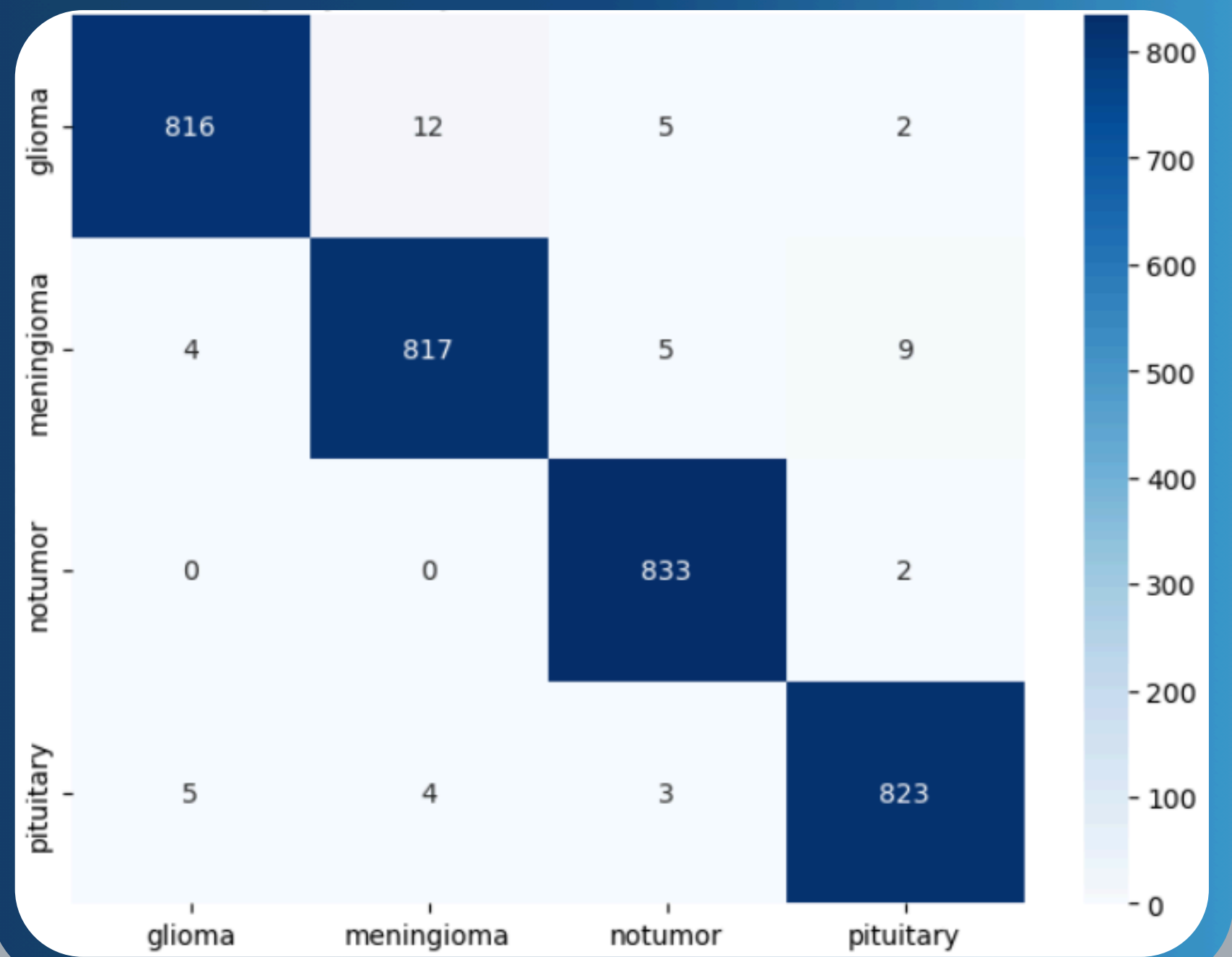
418/418 ————— 303s 725ms/step - accuracy: 0.9867 - loss: 0.0401 - val_accuracy: 0.9832 - val_loss: 0.0668

Training Performance



CNN Model

Confusion Matrix



Classification Report

	precision	recall	f1-score	support
glioma	0.99	0.98	0.98	835
meningioma	0.98	0.98	0.98	835
notumor	0.98	1.00	0.99	835
pituitary	0.98	0.99	0.99	835
accuracy			0.98	3340
macro avg	0.98	0.98	0.98	3340
weighted avg	0.98	0.98	0.98	3340

ResNet50 Model

Training Started

Epoch 1/10

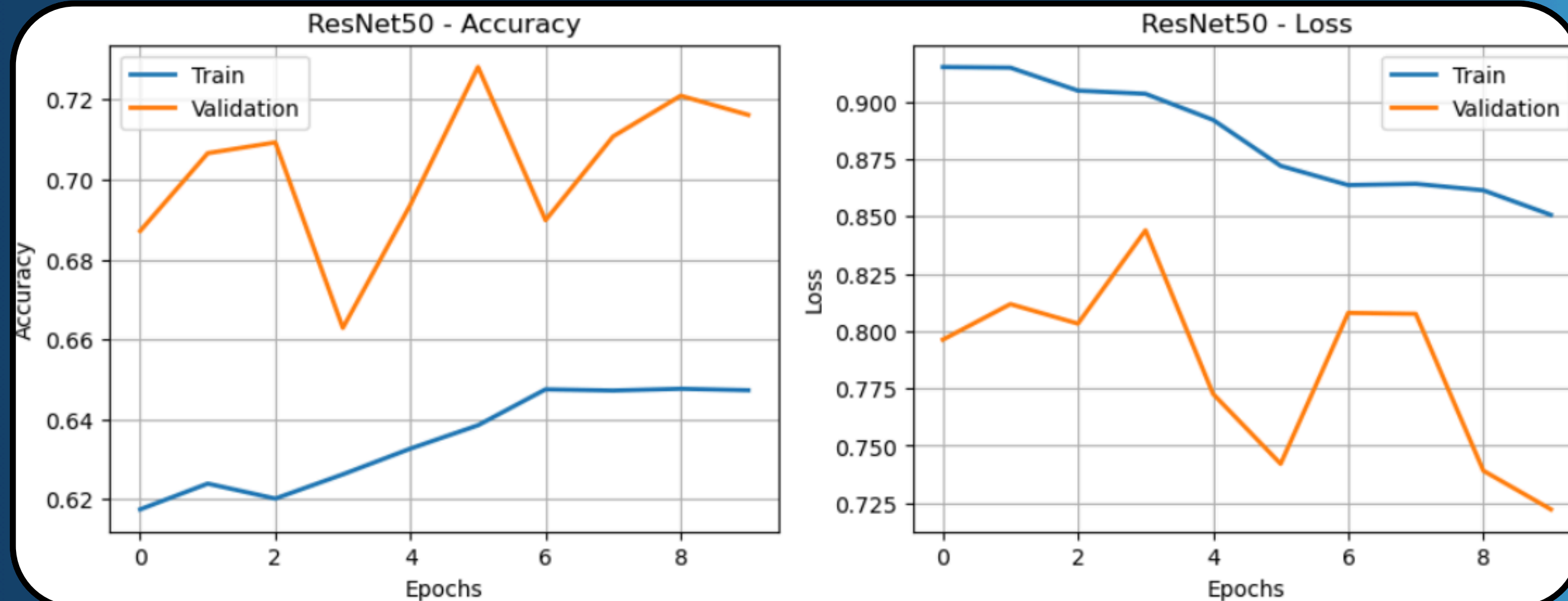
418/418 ————— 657s 2s/step - accuracy: 0.6175 - loss: 0.9211 - val_accuracy: 0.6871 - val_loss: 0.7963

Training Completed

Epoch 10/10

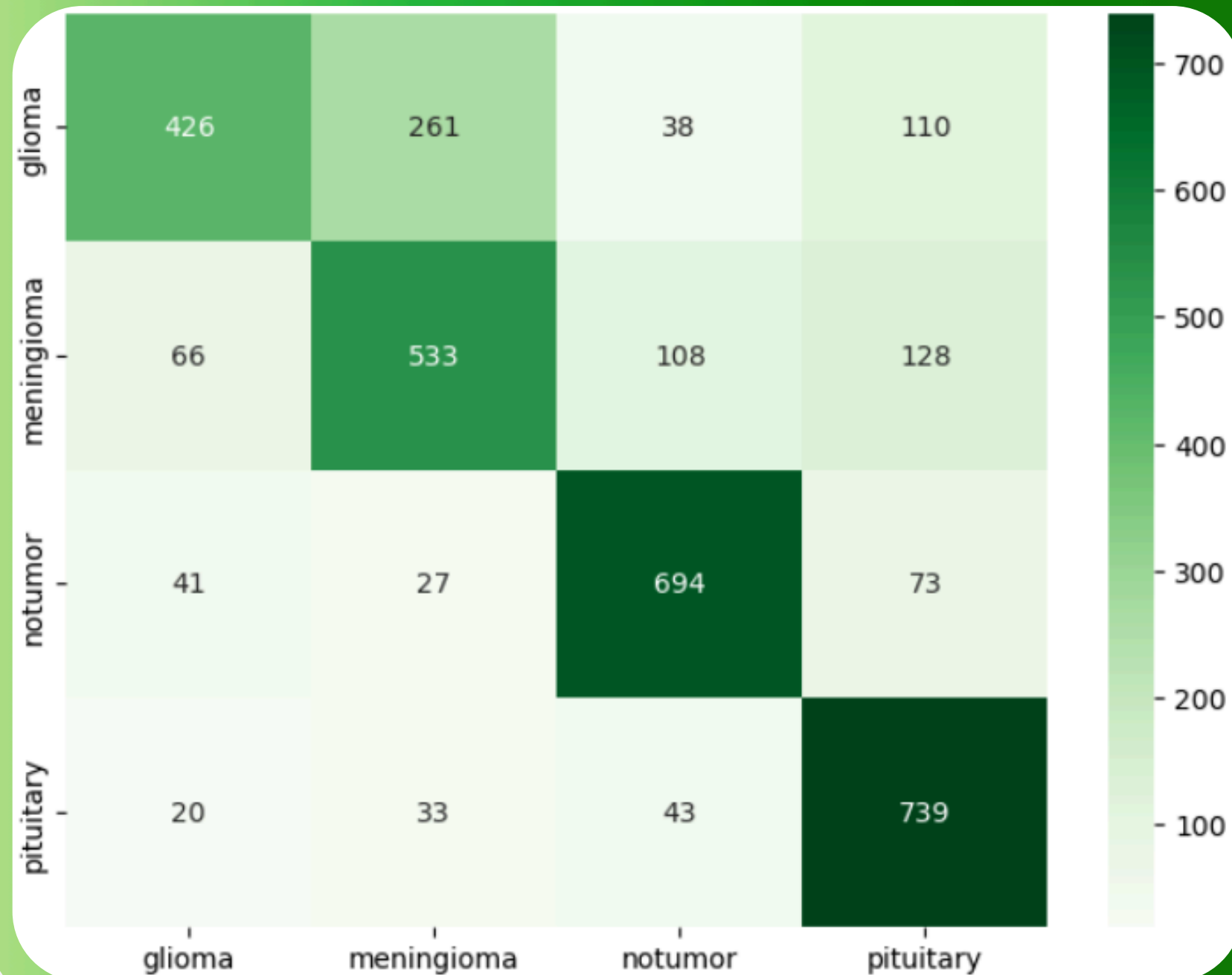
418/418 ————— 874s 2s/step - accuracy: 0.6472 - loss: 0.8572 - val_accuracy: 0.7162 - val_loss: 0.7222

Training Performance



ResNet50 Model

Confusion Matrix



Classification Report

	precision	recall	f1-score	support
glioma	0.77	0.51	0.61	835
meningioma	0.62	0.64	0.63	835
notumor	0.79	0.83	0.81	835
pituitary	0.70	0.89	0.78	835
accuracy			0.72	3340
macro avg	0.72	0.72	0.71	3340
weighted avg	0.72	0.72	0.71	3340

VGG16 model

Training Started

Epoch 1/10

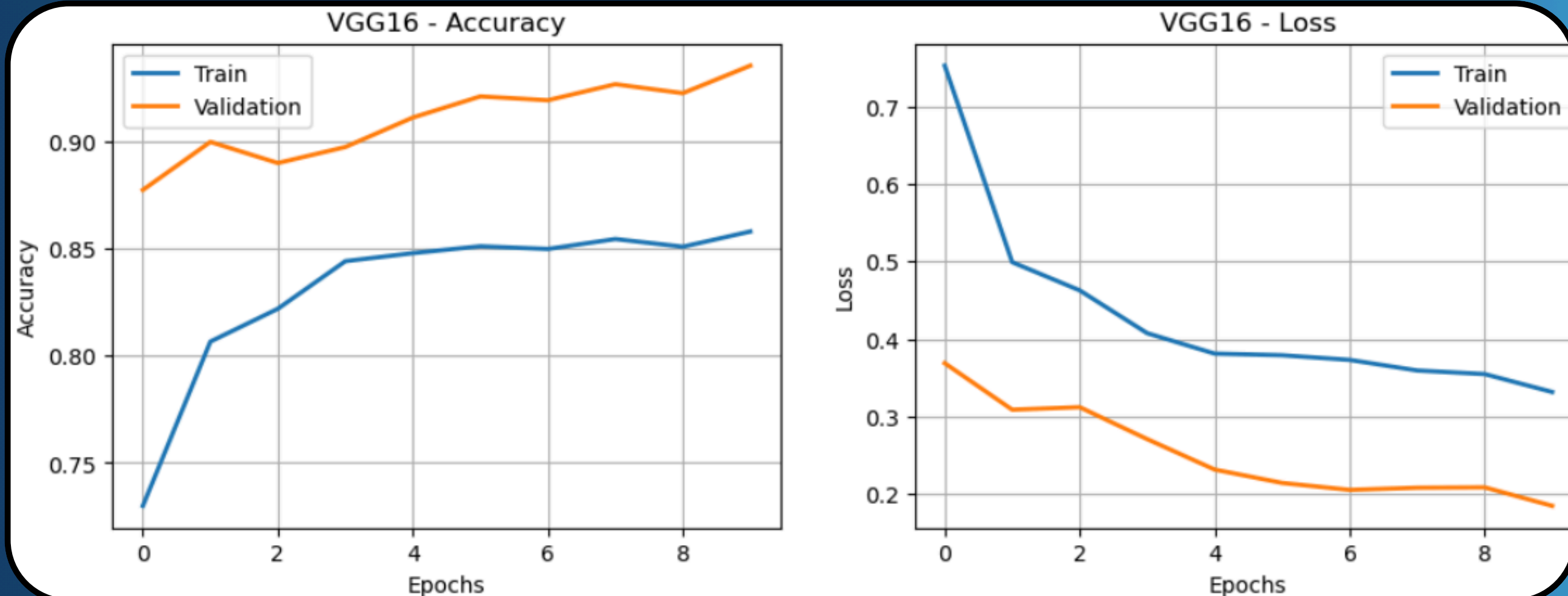
418/418 ————— 2381s 6s/step - accuracy: 0.6484 - loss: 1.2251 - val_accuracy: 0.8775 - val_loss: 0.3686

Training Completed

Epoch 10/10

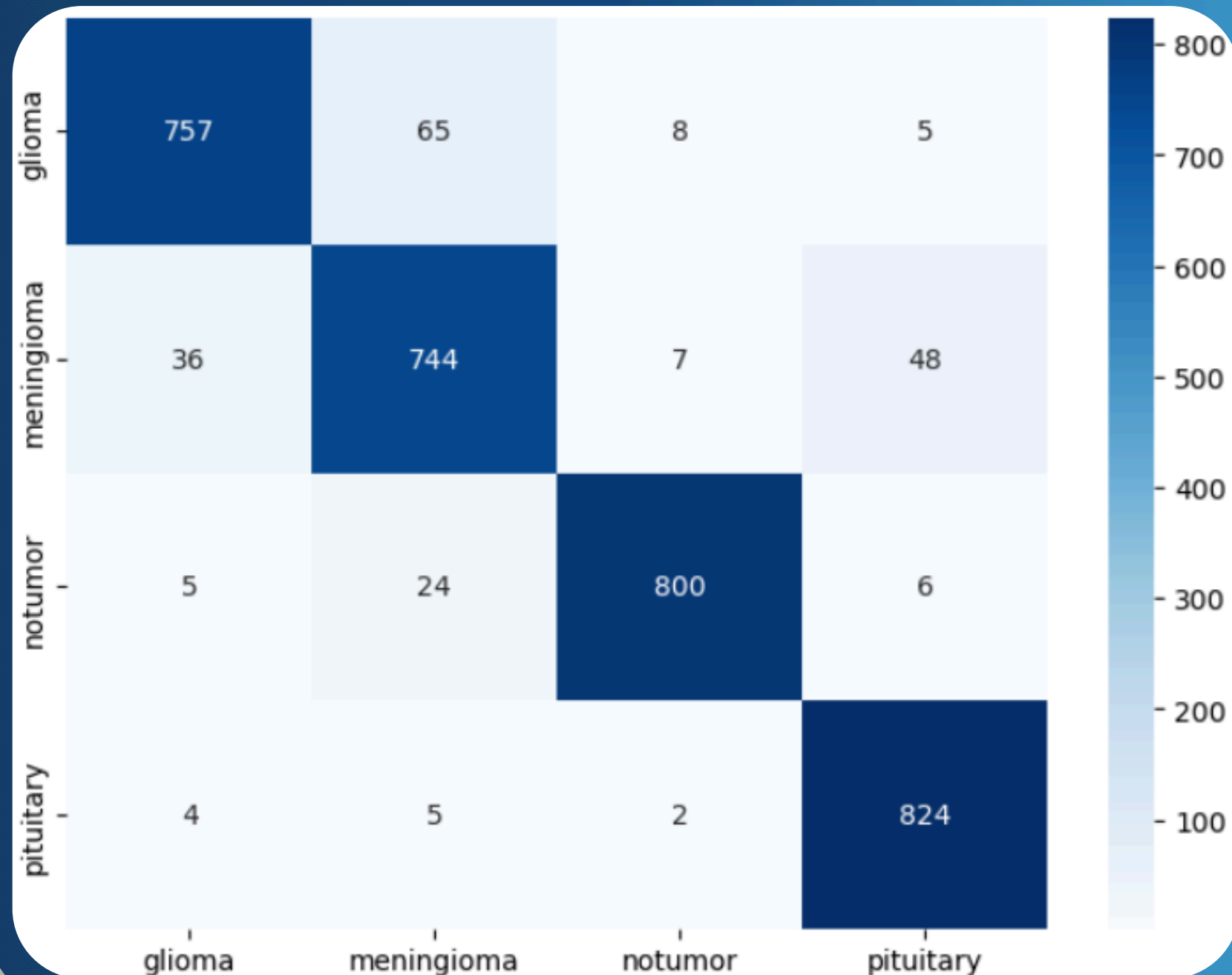
418/418 ————— 2797s 7s/step - accuracy: 0.8641 - loss: 0.3203 - val_accuracy: 0.9356 - val_loss: 0.1845

Training Performance



VGG16 model

Confusion Matrix



Classification Report

	precision	recall	f1-score	support
glioma	0.94	0.91	0.92	835
meningioma	0.89	0.89	0.89	835
notumor	0.98	0.96	0.97	835
pituitary	0.93	0.99	0.96	835
accuracy			0.94	3340
macro avg	0.94	0.94	0.94	3340
weighted avg	0.94	0.94	0.94	3340

Segmentation

LSMAtt-Net Model

LSMAtt-Net is an attention-based model that accurately segments brain tumors by capturing important image features and producing precise tumor boundaries.

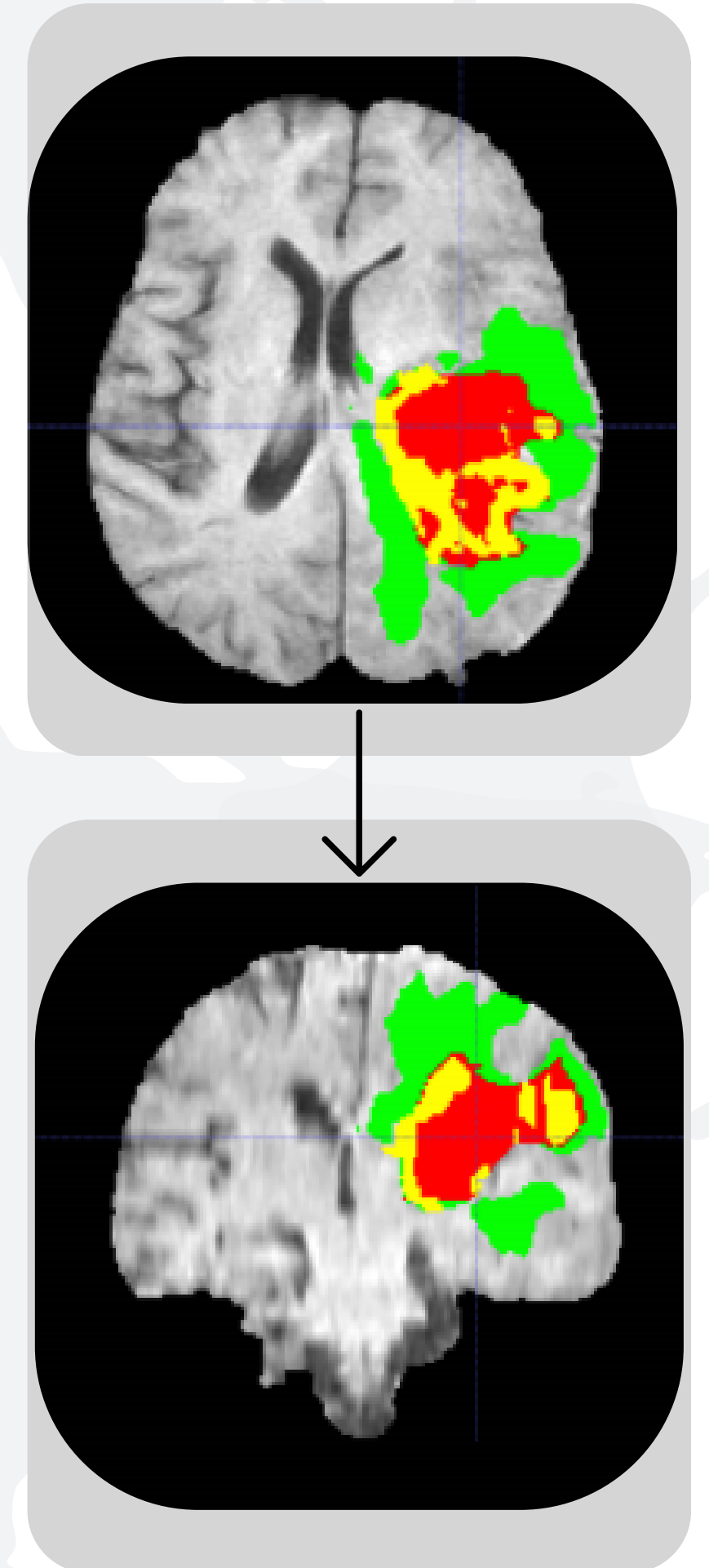
Model Performance

Average
IoU

51.07%

Best IoU

92.40%



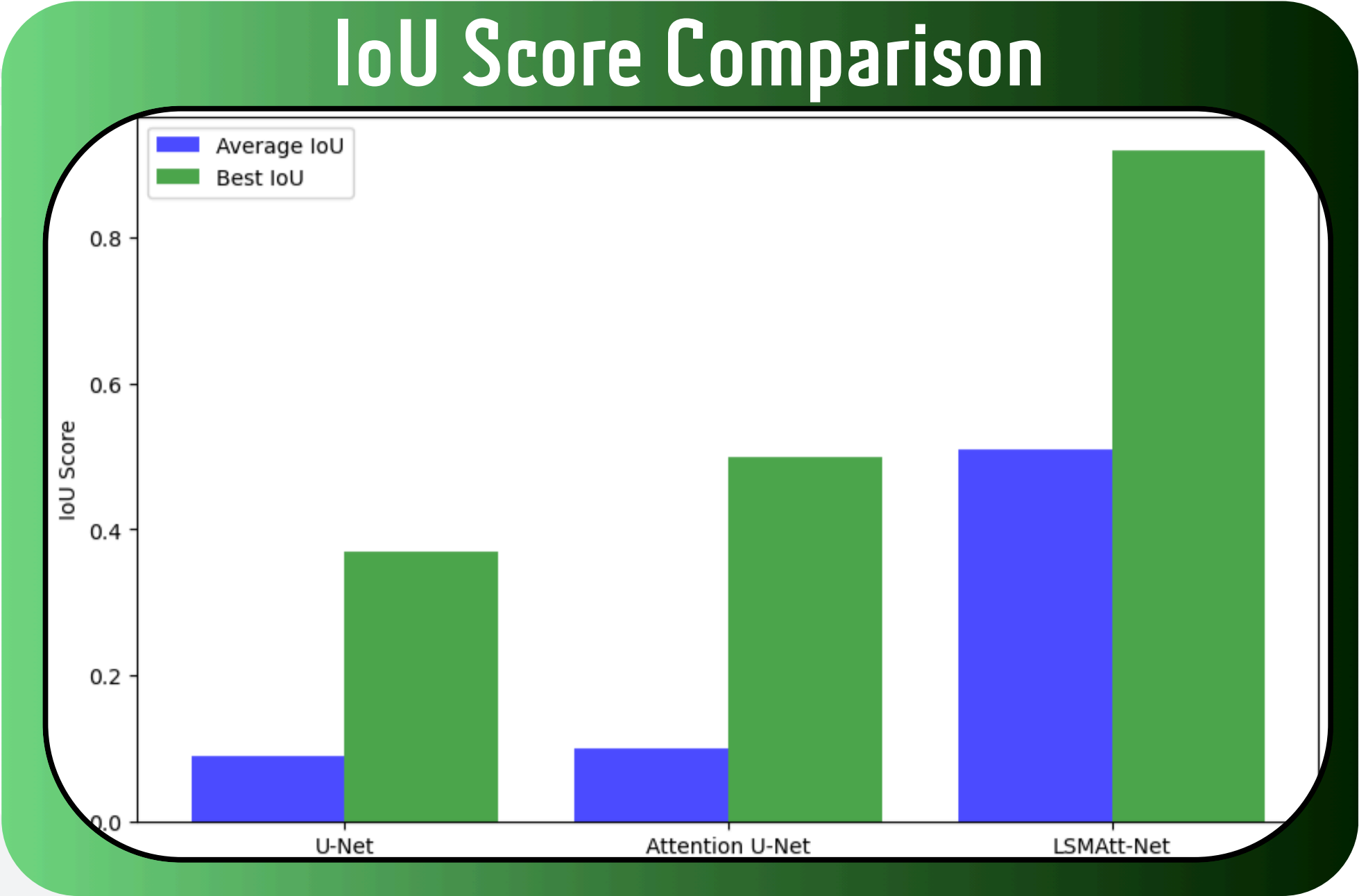
epochs & IoU result

Training Started

Epoch 1/30
492/492 ————— 0s 1s/step - accuracy: 0.8960 - loss: 0.7116
Epoch 1: val_accuracy improved from None to 0.98080, saving model to lsmatt_net

Training Completed

Epoch 30: val_accuracy did not improve from 0.99373
492/492 ————— 485s 986ms/step - accuracy: 0.9973 - loss: 0.0683
Restoring model weights from the end of the best epoch: 30.





System Workflow

1 User

2 Web Application

3 Upload MRI Image

4 Classification (CNN)

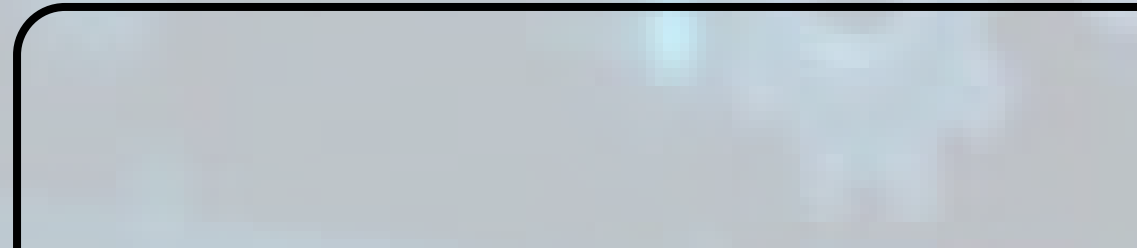
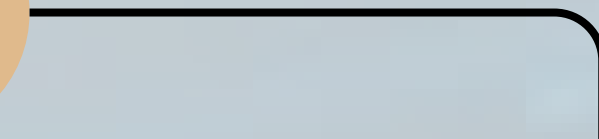
4 Segmentation (LSMAtt-Net)

5 Analysis Results

6 Chatbot

6 Recommendation

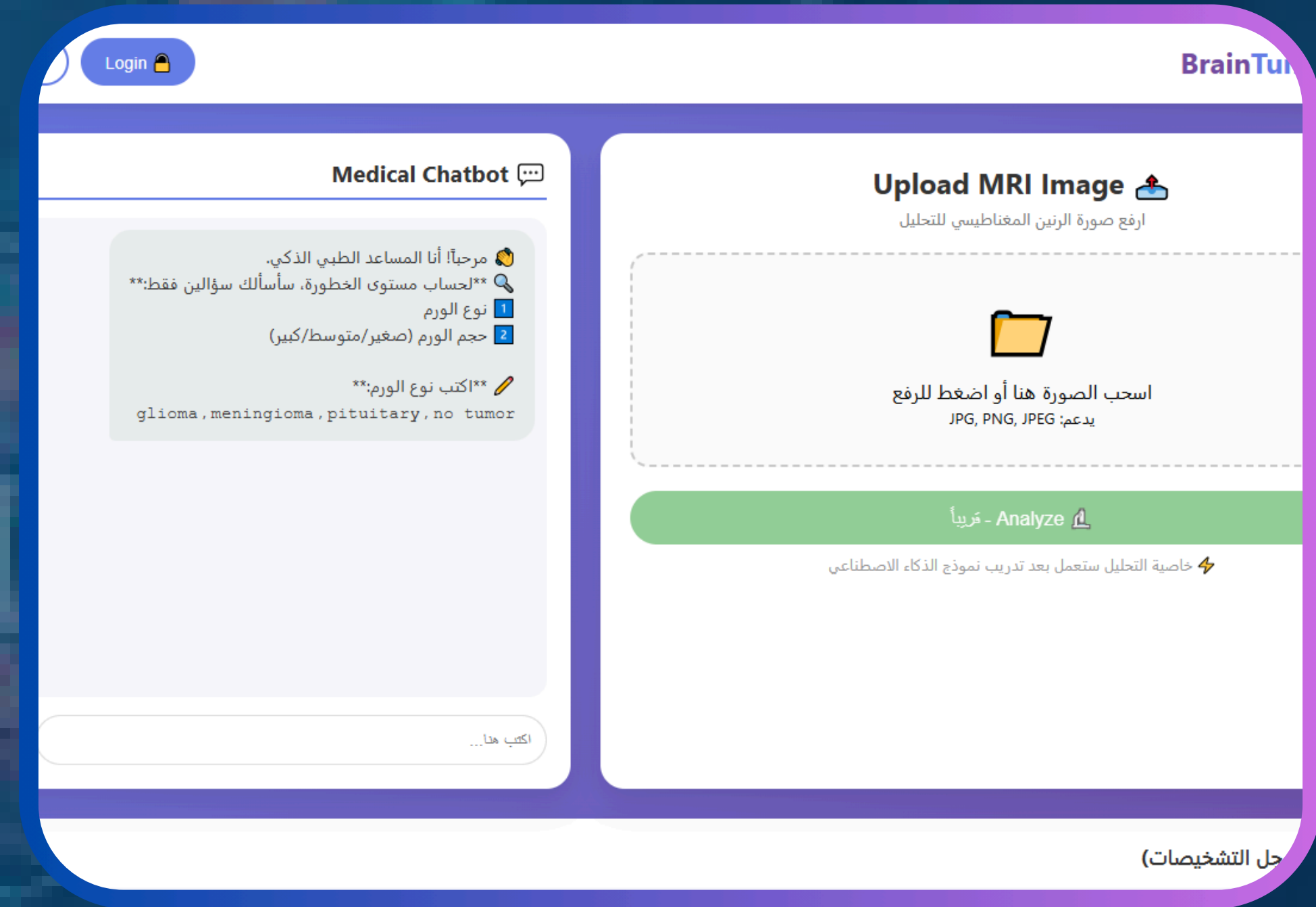
6 PDF Report



User Interface

Main Components

-  MRI Image Upload
-  Analyze Button
-  AI Medical Chatbot
-  Patient History
-  User Authentication



This is the main interface of our web application. It allows users to upload MRI images, analyze them using AI models, interact with the medical chatbot, and view their previous analysis history after logging in.

Conclusion

1.

Successfully classified brain tumors using CNN.

2.

Accurately segmented tumor regions using LSMAtt-Net.

3.

Developed a user-friendly web application.

4.

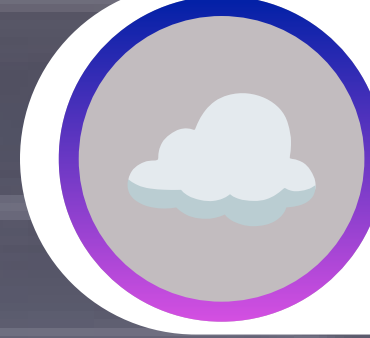
Integrated recommendation, chatbot, and PDF report generation

Future Work



Mobile Application

Develop a mobile version for Android & iOS.



Cloud Deployment

Deploy the system on cloud platforms.



Hospital Integration

Integrate with hospital information systems.



Multi-language Support

Support multiple languages for wider accessibility.



Clinical Validation

Validate the system using real clinical data in collaboration with hospitals.



**Thank
You**